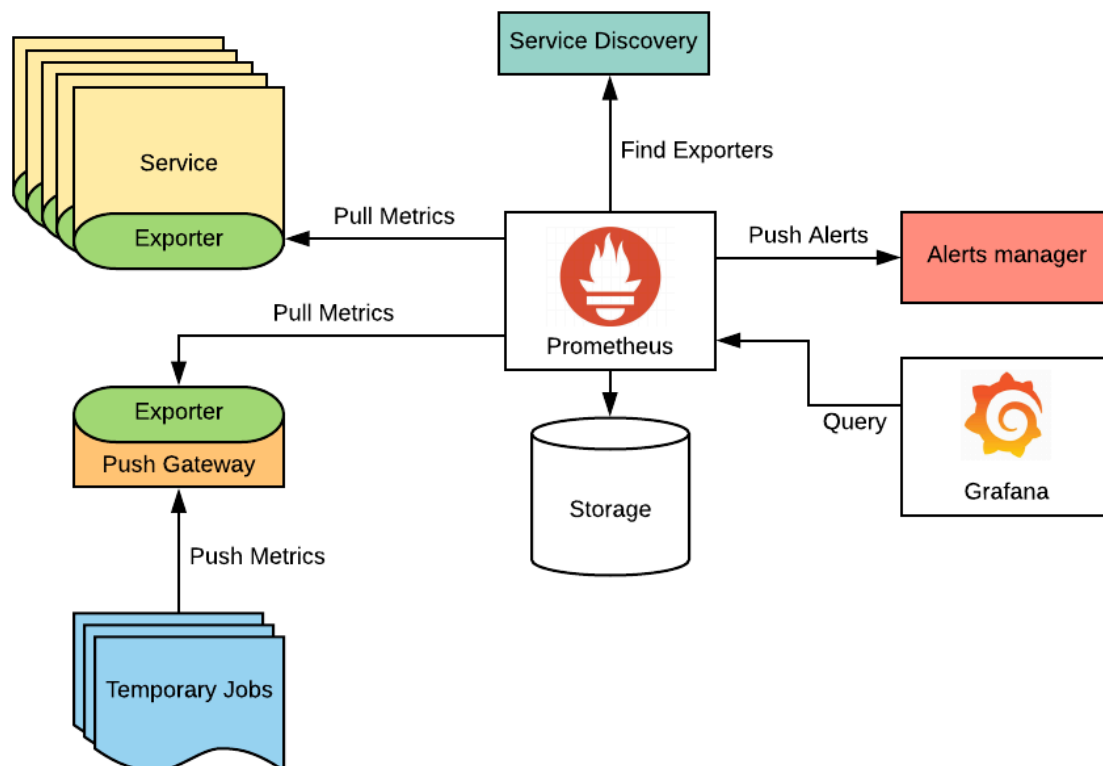


Monitoring Prometheus & Node Exporter On Linux

Prometheus is an open-source systems monitoring and alerting toolkit originally built at SoundCloud. Since its inception in 2012, many companies and organizations have adopted Prometheus, and the project has a very active developer and user community. It is now a standalone open source project and maintained independently of any company. To emphasize this, and to clarify the project's governance structure, Prometheus joined the Cloud Native Computing Foundation in 2016 as the second hosted project, after Kubernetes.

Prometheus collects and stores its metrics as time series data, i.e. metrics information is stored with the timestamp at which it was recorded, alongside optional key-value pairs called labels.



Tasks:

1. Install Prometheus (server)
2. Configure prometheus
3. Create Prometheus Systemd Service
4. Access Prometheus Web UI
5. Install Node Exporter
6. Configure Node Exporter
7. Configure the Server as Target on Prometheus Server

STEP 1 :

Install Prometheus (server) :

Setup Prometheus Binaries :

Update the apt package repositories :

```
sudo apt update -y
```

Downloading Prometheus Binary :

```
cd /tmp
```

```
curl -LO
```

```
https://github.com/prometheus/prometheus/releases/download/v2.37.0/prometheus-2.37.0.linux-amd64.tar.gz
```

Create Prometheus user :

```
sudo useradd --no-create-home --shell /bin/false prometheus
```

Extract the binaries to executable path :

extract the downloaded file and move the binaries to executable path. We are moving the prometheus and promtool in the commands below:

```
tar -xzf prometheus-2.37.0.linux-amd64.tar.gz
```

```
sudo mv prometheus-2.37.0.linux-amd64/prometheus /usr/local/bin/
```

```
sudo mv prometheus-2.37.0.linux-amd64/promtool /usr/local/bin/
```

Ensure prometheus user owns the binaries using linux chown command:

```
sudo chown prometheus:prometheus /usr/local/bin/prometheus
```

```
sudo chown prometheus:prometheus /usr/local/bin/promtool
```

Create required directories and copy configs :

```
sudo mkdir /etc/prometheus
```

```
sudo mkdir /var/lib/prometheus
```

```
sudo chown prometheus:prometheus /etc/prometheus
```

```
sudo chown prometheus:prometheus /var/lib/prometheus
```

Move the consoles and console_libraries directories from prometheus-files to /etc/prometheus folder and change the ownership to prometheus user :

```
sudo mv prometheus-2.37.0.linux-amd64/consoles /etc/prometheus
```

```
sudo mv prometheus-2.37.0.linux-amd64/console_libraries /etc/prometheus
```

```
sudo chown -R prometheus:prometheus /etc/prometheus
```

STEP 2 :

Setup prometheus Configuration:

All the prometheus configurations should be present in /etc/prometheus/prometheus.yml file. Open the file:

```
sudo nano /etc/prometheus/prometheus.yml
```

Add this configuration to the file:

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
```

Change the ownership of the file to prometheus user :

```
sudo chown prometheus:prometheus /etc/prometheus/prometheus.yml
```

STEP 3 :

Create Prometheus Systemd Service

Create Prometheus Systemd Service File :

We will use this service to manage the service starting and stopping :

```
sudo nano /etc/systemd/system/prometheus.service
```

Add this configuration to the file:

```
[Unit]
Description=Prometheus
Wants=network-online.target
After=network-online.target

[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus \
--config.file /etc/prometheus/prometheus.yml \
--storage.tsdb.path /var/lib/prometheus/ \
--web.console.templates=/etc/prometheus/consoles \
--web.console.libraries=/etc/prometheus/console_libraries

[Install]
WantedBy=multi-user.target
```

Reload the systemd service to register the prometheus service and start the prometheus service :

```
sudo systemctl daemon-reload
```

```
sudo systemctl start prometheus
```

Check the prometheus service status :

```
sudo systemctl status prometheus
```

STEP 4 :

Access Prometheus Web UI :

Now you will be able to access the prometheus UI on 9090 port of the prometheus server :

<http://<prometheus-server-ip>:9090/graph>

STEP 5 :

Install Node Exporter :

Node exporter is the best way to collect all the Linux server related metrics and statistics for monitoring.

Setup Node Exporter Binary :

Download the latest node exporter package

```
cd /tmp
```

```
curl -LO
```

```
https://github.com/prometheus/node_exporter/releases/download/v1.3.1/node_exporter-1.3.1.linux-amd64.tar.gz
```

Extract the binaries to executable path :

```
tar -xvf node_exporter-1.3.1.linux-amd64.tar.gz
```

Move the node export binary to /usr/local/bin :

```
sudo mv node_exporter-1.3.1.linux-amd64/node_exporter /usr/local/bin/
```

STEP 6 :

Configure Node Exporter :

Create a Custom Node Exporter Service :

Create a node_exporter user :

```
sudo useradd -rs /bin/false node_exporter
```

Create a node_exporter service file :

```
sudo nano /etc/systemd/system/node_exporter.service
```

Add this configuration to the file:

```
[Unit]
```

Description=Node Exporter
After=network.target

[Service]
User=node_exporter
Group=node_exporter
Type=simple
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=multi-user.target

Reload the system daemon and star the node exporter service :

```
sudo systemctl daemon-reload  
sudo systemctl start node_exporter  
sudo systemctl enable node_exporter
```

check node exporter :

```
sudo systemctl status node_exporter
```

node exporter would be exporting metrics on port 9100 :

<http://<Node-IP>:9100/metrics>

STEP 7 :

Configure the Server as Target on Prometheus Server :

Server : prometheus

open the prometheus.yml file :

```
sudo nano /etc/prometheus/prometheus.yml
```

Under the scrape config section add the node exporter target as shown below. Change **Node-IP-node-exporter** with your server IP where you have set up node exporter.

In terms of indent , the desired job should be the level of other jobs.

```
- job_name: 'node_exporter_metrics'  
  scrape_interval: 5s  
  static_configs:  
    - targets: [x.x.5.145:9100']
```

Restart the prometheus service for the configuration changes to take place :

```
sudo systemctl restart prometheus
```


Check the target in prometheus web UI :

<http://<prometheus-IP>:9090/targets>

PrometheusAlertsGraphStatus▼Help

Targets

AllUnhealthy

node_exporter_metrics (1/1 up)show less

Endpoint	State	Labels	Last Scrape
http://10.142.0.3:9100/metrics	UP	instance="10.142.0.3:9100"	42ms ago

prometheus (1/1 up)show less

Endpoint	State	Labels	Last Scrape
http://localhost:9090/metrics	UP	instance="localhost:9090"	1.307s ago

Successful

END